

ALLAN KAGIMU SSEBATTA

Software Development | Cybersecurity | Networking | IoT & Embedded Systems
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PROFESSIONAL SUMMARY

Computer Science graduate from Uganda Christian University with a First Class Honours degree. Experienced in software development, cybersecurity, networking, and IoT/embedded systems through internships and independent projects. Completed a placement at the Uganda Industrial Research Institute (UIRI) and holds a First Position award from the HACK UCU 2025 Hackathon. Ready to bring strong technical skills and a practical, results-driven approach to a full-time professional role.

WORK EXPERIENCE

Embedded Systems, IoT & Backend Development Intern — *Uganda Industrial Research Institute, Namanve* May 2024 – Jul 2024

- Designed and tested IoT prototypes using Arduino and NodeMCU, integrating smoke, ultrasonic, and LDR sensors with buzzers, servo motors, and LCD displays.
- Implemented secure user authentication logic and managed structured databases using MySQL for reliable data storage and retrieval.
- Designed system architecture flowcharts and logical diagrams, clarifying data flow and hardware-software interaction across projects.
- Conducted debugging and hardware-software integration testing, improving prototype reliability ahead of demonstration deadlines.
- Collaborated with a cross-functional team of engineers and interns on research-driven automation and smart system projects.

Computer Maintenance & Repair Intern — *Vision Africa, Kayunga* May 2024

- Installed and configured hardware and software components to restore and optimise workstation performance.
- Produced technical documentation covering system configurations, troubleshooting procedures, and user guides for staff reference.
- Diagnosed and resolved hardware and software faults using structured repair trees, reducing downtime for end users.

Computer Maintenance & Repair Intern — *Seroma Christian High School, Mukono* Feb 2024

- Maintained and repaired school computer lab equipment, ensuring optimal performance for students and staff.
- Created and maintained clear documentation of system configurations and troubleshooting steps for future reference.
- Organised and labelled cables and devices, improving workstation tidiness and ease of maintenance.

KEY PROJECTS

Vanguard — Network Intrusion Detection System (NIDS) — *Cybersecurity | Architecture & Design*

Designed the full system architecture for a NIDS including traffic monitoring logic, threat detection workflows, and automated alert generation using structured flow diagrams.

IoT Intrusion Detection & Alert System — *Arduino | NodeMCU | Blynk | IoT*

Built a real-time smoke detection system with ultrasonic sensors, buzzer alerts, LED indicators, servo control, and LCD display. Integrated Blynk for live mobile dashboard monitoring and instant alerting.

Edge-Enabled Smart Irrigation System — *IoT | Edge Computing | Embedded Systems*

Developed an IoT irrigation system for smallholder farmers using edge computing and real-time sensor monitoring to automate water usage and improve farming efficiency.

Secure Authentication System — *Python | Backend | Security*

Implemented a backend authentication system with registration and login functionality, password hashing, JSON-based credential storage, and structured input validation in Python.

TECHNICAL SKILLS

Languages: Python, C, C++, JavaScript, HTML, CSS
Embedded & IoT: Arduino, NodeMCU, sensor integration, Blynk, edge computing
Web Development: Frontend (HTML/CSS/JS), Backend (Python), REST APIs
Databases: MySQL, JSON-based storage
Cybersecurity: NIDS design, digital forensics, authentication systems
Networking: Local Area Networking, TCP/IP fundamentals
Tools & Methods: System architecture design, flowcharting, debugging, documentation

EDUCATION

BSc Computer Science (First Class Honours) — <i>Uganda Christian University, Mukono</i>	Sep 2023 – Jul 2026
Uganda Advanced Certificate of Education (A-Level) — <i>St. Julian High School, Gayaza</i>	Feb 2020 – Dec 2022
Uganda Certificate of Education (O-Level) — <i>Setlight Quality Secondary School, Entebbe</i>	Feb 2016 – Dec 2019

ACHIEVEMENTS & ACTIVITIES

- First Class Honours — BSc Computer Science, Uganda Christian University (2026)
- Certificate of Excellence — First Position, HACK UCU 2025 Hackathon, Uganda Christian University
- Participant in research-driven IoT and automation projects at Uganda Industrial Research Institute
- CCNA: Enterprise Networking, Security, and Automation Certificate
- Successfully completed practical training and achieved a technician level competence

ADDITIONAL INFORMATION

Languages: English (Fluent), Luganda (Native)

References

Dr. Innocent Ndibatya — Head, Computing & Technology Dept, Uganda Christian University | indibatya@ucu.ac.ug
Mr. Kasole Ahmed — Lecturer, Uganda Christian University | akasole@ucu.ac.ug | +256 780 573 769